Dachuan Song

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Google Scholar | GitHub | LinkedIn | Personal Website

Education

George Mason University, Fairfax, Virginia, USA — Jan 2024 – Present
College of Engineering and Computing
Ph.D. in Electrical and Computer Engineering, Current GPA: 3.91

Advisor: Prof. Xuan Wang

- **Major Courses:** *Linear Systems, Random Processes, Kalman Filtering, Distributed Control and Optimization, System Identification, Advanced learning from data*

University of Southampton, Southampton, United Kingdom — Sept 2022 – Sept 2023
School of Electronics and Computer Science
M.Sc. in Computer Science, GPA: 3.8 (Top 3%)

- **Major Courses:** Advanced Machine Learning, Advanced Deep Learning, Computer Vision, Bayesian, Active & Reinforcement Learning

Xinjiang University, Xinjiang, China — Sept 2017 – Jun 2021
School of Software
B.Eng. in Software Engineering, GPA: 3.4 (Top 5%)

Publications

Conference Papers and Preprints:

Elastic Spectral State Space Models for Train-Once Budgeted Inference — 2026
Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang
https://arxiv.org/abs/2601.22488
*Accepted to **NeurIPS 2026** Main Track (Poster); proceedings forthcoming. arXiv:2601.22488, 2026.*

Elastic Selective Spectral Hybrids for Train-Once, Export-Many Budgeted Inference — Sep 2026
Dachuan Song, Chuchu Chen, Xuan Wang
https://arxiv.org/abs/2609.32486

Mergeable Model-Side Aggregation States for Long-Context Language Models — Jul 2026
Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang
https://arxiv.org/abs/2607.26448

Causality-based Subject and Task Fingerprints using fMRI Time-series Data — Nov 2024
Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang
https://doi.org/10.1145/3698587.3701342
ACM BCB '24, Article 13, pp. 1–10 (Oral Presentation).

Journal Articles:

Reconstructing brain causal dynamics for subject and task fingerprints using fMRI time-series data — Oct 2025
Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang
https://doi.org/10.1007/s13755-025-00388-w
Health Information Science and Systems 13, Article 70 (2025).

Research Projects

State-Space Language Models for Long Context Memory and Reasoning Sep 2026 – Present

- Developing language models based on state-space architectures for reliable long-context factual recall and multi-hop reasoning with linear-time sequence processing.
- Aiming to support consistent, evidence-grounded decisions in language-model agents and future language-guided robotics while preserving general language quality and keeping inference costs low.
- Designing matched comparisons with Transformer and state-space baselines, controlled ablations and real-text question-answering evaluations; assessing long-context generalization, perplexity, training throughput, memory use and decoding costs.

Elastic Selective Spectral Hybrids for Budgeted Inference Jul 2026 – Sep 2026

- Gave each spectral channel a recurrent state with input-dependent decay and write/read gates, so it can retain relevant context and forget distractions. Sliding-window attention handles recent tokens.
- Trained channel count and feed-forward width as capacity tiers, producing compact exports from one checkpoint. At 1.53B parameters, fused batch-one decoding took 1.37 ms/token on a B300, 2.14–2.80$\times$ faster than the tested Mamba-2 and Mamba-3 implementations at matched parameter counts.

Elastic Spectral State Space Models for Train-Once Budgeted Inference Jan 2026 – May 2026

- One training run yields independently deployable models for different budgets. Direct truncation removes ordered Hankel channels and their parameters from each smaller export.
- Input-adaptive gates tune channel contributions; budget dropout trains every retained prefix. Across language, audio, long-range, and offline reinforcement-learning tasks, quality changes smoothly as the budget shrinks.

Mergeable Model-Side Aggregation States for Long-Context Language Models Jul 2026 – Aug 2026

- Added a 2 KiB mergeable HyperLogLog state beside a frozen language model to track distinct counts and set evidence across context chunks.
- Reached 99.2% accuracy on 3,969 aggregate-then-reason tasks with Gemma 4 (31B), without rereading the full history.

Machine Learning for Brain Activity Patterns using EEG (Master's Project) Apr 2023 – Oct 2023

- Combined CNN spatial features with LSTM temporal modeling for EEG brain-pattern recognition.
- Used autoencoder pretraining, attention, and transfer learning to extract features from limited EEG data.

Internship Experience

Machine Learning Engineer Intern, Optosurgical May 2026 – Aug 2026

- Recognized surgical phases and instrument actions in video, providing procedure-state signals to downstream systems.
- Compressed and fine-tuned Depth Anything 3 to 29% of its original size while retaining near-full depth-estimation performance.

Technical Skills

- **Architectures & Inference:** spectral and selective state space models, recurrent memory, sliding-window attention, long-context aggregation, multi-budget model export.
- **ML Engineering:** PyTorch, CUDA, FFT convolution, multi-GPU training, model compression, fused recurrent decoding, reproducible evaluation.
- **Modeling & Control:** linear systems, Kalman filtering, system identification, LQR/MPC, autoregressive models.